

Spacetime from Cost

A functor from cost-accounted ciGSLTs to measured causal sets

Companion note to “Continued Interactive GSLTs and the Cost Endofunctor”

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Abstract

The cost endofunctor $\mathfrak{C}$ of [1] equips an interactive graph-structured lambda theory with two combination operators of opposite character: a spatial contact constructor K , recording what is near what, and a temporal cons $::$ on located resource stacks, recording what is spent next. That note observed, in its closing correspondences, that when K carries equations the surface at which a cut fires must be carried nominally by a nearness operator, and that the resulting *located* stack $S(I, -)$ ties a temporal resource to a spatial surface, so that authority entangles space and time into “a spacetime.” Here we make that map precise. We define a functor $\mathcal{M}$ from pointed cost-accounted continued interactive GSLTs to *measured causal sets* — locally finite partial orders with the counting measure, the objects that causal-set quantum gravity takes as discrete spacetimes via Sorkin’s slogan *Order + Number = Geometry* [2, 3]. Events are cut firings; the causal order is enabling; concurrency is Bernstein independence of footprints, while all nondeterminism (races) is carried by conflict; the measure is spent phlogiston. We prove: that every history of $\mathfrak{C}(G)$ is a causal set (local finiteness); that the counting measure of an event’s causal past equals the phlogiston spent to reach it (Number = Volume), via the modulus proposition of [1]; that discrete proper time is the longest chain and satisfies the Lorentzian reverse triangle inequality; that a single located stack contributes a timelike chain while non-near surfaces contribute spacelike antichains, so that located authority is exactly what forces a $(1+n)$ ordering fraction rather than a degenerate chain or antichain; that the construction is functorial and that the measured causal set is an invariant of *hereditary history-preserving* bisimulation (and, pointedly, *not* of ordinary interleaving bisimulation, which forgets the concurrency that is the space); and that one reduction step is one cover in the configuration domain whose new links are a local functional of the consumed footprint — the discrete shape of the field-equation coupling. The continuum Lorentzian manifold appears only as the Hauptvermutung class of a manifoldlike causal set, the fourth and idealizing quotient. Throughout we separate what is proved here from the standard external theorems we invoke.

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*lgreg.meredith@f1r3fly.io. This note was developed in dialogue with Claude (Anthropic); the framing of located authority as the Lorentzian welding of K and $::$, and the identification of the spacetime invariant with hereditary history-preserving bisimulation, emerged in that exchange.

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1 Introduction

The note [1] closed with two correspondences to physical theories, offered in the spirit of Curry–Howard as resemblances detailed enough to invite investigation. The first read a K -tree as a distribution of information, cut elimination as a dynamics that changes adjacency, and the coupling between the two — the information distribution shaping the nearness structure, the nearness structure selecting the next redistribution — as the shape of Einstein’s intuition. It then observed that object capabilities sharpen rather than dissolve the picture: a *located* resource stack $S(I, -)$ ties a temporal resource (what may be spent next) to a spatial surface (who is near), so a draw requires both spatial proximity ($\text{near}(I, J)$ defined) and temporal availability (a token at the head of the stack). Space and time can then no longer be varied independently; authority *entangles* them, and “what was a product of a space and a time becomes a single coupled structure gating interaction — a spacetime.”

That note left the formal development to future work. This note supplies it, in the one direction that is both precise and already standard: the target is not a manifold but a *causal set*. The causal-set programme of Bombelli, Lee, Meyer, and Sorkin [2] takes a locally finite partial order, equipped with the counting measure, as a discrete spacetime in its own right, recovering the continuum only as an approximating limit. Its founding slogan, *Order + Number = Geometry* [3], is exactly what the two monoids of $\mathfrak{C}$ supply: the causal order is the enabling order of cut firings, and the number — the volume — is the spent phlogiston. A Lorentzian manifold, when it exists, is then the Hauptvermutung class of the causal set: the family of manifolds into which it faithfully embeds, unique up to approximate isometry. This is the precise content of the user’s “(quotiented) spacetime manifold,” and it factors the *quotiented* into four named identifications (Section 10).

What is proved and what is cited. We are explicit about the boundary. Proved here, from the operational semantics of $\mathfrak{C}(G)$ and the structural results of [1], are: the diamond/locality of

independent firings (Lemma 3.5); the prime-event-structure axioms (Lemma 4.4); local finiteness, i.e. that every history is a causal set (Theorem 5.2); Number = Volume = phlogiston (Theorem 6.2); the Lorentzian reverse triangle inequality for proper time (Proposition 7.2); the signature criterion relating located authority to the ordering fraction (Proposition 7.3); functoriality and the HHP-bisimulation invariance of the spacetime (Theorem 8.5, Corollary 8.6); and the naturality of the cover along reduction (Proposition 9.1). Cited as standard external results, not reproved, are: the representation of asynchronous transition systems and trace languages by prime event structures and prime algebraic domains [8, 9, 10, 11, 12]; the characterisation of event-structure isomorphism by hereditary history-preserving bisimulation [13]; the Myrheim–Meyer dimension estimator [4, 5]; and the closeness/Hauptvermutung theory of causal sets and Lorentzian manifolds [6, 7]. We use the external results only to package and to interpret; every claim that depends on the specifics of $\mathfrak{C}$ is proved.

2 Recollection: the cost-accounted theory

We recall just enough of [1] to be self-contained, fixing the notation this note operates on. Let G be a continued interactive GSLT: an interactive theory presented as an *interaction cut*. Its dynamics brings two introductions into contact along matching interaction surfaces and contracts them,

$$K(K_p(I, C_p), K_e(J, C_e)) \longrightarrow \text{compute}(I, J, C_p, C_e), \quad (1)$$

where K is the contact constructor, $K_p(I, C_p)$ and $K_e(J, C_e)$ the program- and environment-side introductions, I, J their surfaces, and compute Milner’s pseudo-application.

The cost endofunctor $\mathfrak{C}$ produces from G a metered calculus $\mathfrak{C}(G)$: it signs terms with elements of a commutative signature monoid $(\text{Sig}, *, ())$ via $\# = \text{digest} \circ \text{cf}$, adjoins a *temporal token stack*, makes K polymorphic, and replaces the cut by a *gated* family of rules. The temporal stack is a free monoid (a list) under cons $::$; it is never commutative, because reduction is sequential. Following Section 11 of [1], when K carries any equation the stack is *located* at the surface that may draw on it, written

$$S(I, s :: S'),$$

a stack indexed by the surface I , with head token s and tail S' . The metered reduction is the gated cut: schematically, writing $\mathcal{C}[-]$ for a configuration context,

$$\mathcal{C}\left[K(K_p(I, C_p), K_e(J, C_e)); S(L, s :: S')\right] \longrightarrow \mathcal{C}\left[\text{compute}(I, J, C_p, C_e); S(L, S')\right], \quad (2)$$

which fires only when $\text{near}(I, J) = L$ is defined and the head token s matches the signature guard. Firing the rule is *forcing*: it unwraps the metered thunk by spending exactly one head token. Structural congruence $\equiv$ (which includes the equations of K and the location-respecting rearrangements of the bag) consumes no token and leaves the stack invariant. We import the two structural results of [1] that we shall use.

Property 2.1 (No leak / wrapping is a sorting invariant; [1, Lemma on subject reduction]). *In $\mathfrak{C}(G)$ every redex is born inside a wrapper, and reduction preserves wrapping: the only run-time interaction with tokens is unwrapping. In particular a firing modifies no token other than the head it spends, and creates no enabling for another redex except through the contractum it produces.*

Property 2.2 (Modulus; [1, Prop. “stack consumption is the modulus”]). *For a terminating reduction in $\mathfrak{C}(G)$ the number of stack cells consumed equals the number of forced redexes, which*

equals the number of cuts eliminated — including those introduced by duplication, each charged when forced. The consumed prefix of the temporal stack is an exact operational modulus for the cut elimination.

From the shape of (2) and Property 2.1 we extract the single locality fact on which the concurrency analysis turns. It is a property of the rule, recorded here as the working hypothesis we verify against [1] rather than reprove.

Property 2.3 (Locality of forcing). *A firing $C[\rho] \longrightarrow C[\rho']$ that contracts the cut instance ρ and pops the head of the located stack at L leaves every subterm of $C[\rho]$ disjoint from ρ and from that head cell unchanged, up to $\equiv$.*

This is immediate from (2): the left side names exactly the contracted cut and the spent head cell, and Property 2.1 forbids any ambient side effect, so the rewrite is confined to the named material.

3 The pointed domain and firings

A causal set needs an origin: the spacetime of a computation is the spacetime of a run, so we point the objects with an initial configuration.

Definition 3.1 (Pointed cost-accounted theory). The category **Cost-ciGSLT**_• has as objects pairs $(\mathfrak{C}(G), t_0)$ with G a continued interactive GSLT and t_0 a closed configuration of $\mathfrak{C}(G)$ (a wrapped, signed term carrying located stacks). A morphism $(\mathfrak{C}(G), t_0) \rightarrow (\mathfrak{C}(G'), t'_0)$ is a **ciGSLT**-morphism $f : \mathfrak{C}(G) \rightarrow \mathfrak{C}(G')$ in the sense of [1] (bisimulation-preserving and quote-faithful) with $f(t_0) = t'_0$. We write $\text{Reach}(t_0)$ for the set of configurations reachable from t_0 under $\longrightarrow$, taken modulo $\equiv$ (Property 2.3 makes $\longrightarrow$ well defined on $\equiv$ -classes; see Lemma 3.3).

Definition 3.2 (Redex occurrence, firing, footprint). A *redex occurrence* in a configuration t is the data $r = (C, \kappa)$ of a context together with a cut instance $\kappa = (I, J, C_p, C_e; L, s)$ filling its hole, to which the gated rule (2) applies: $\text{near}(I, J) = L$ and the head of the stack at L is the matching token s . We write $t \xrightarrow{r} t'$ for the corresponding *firing*. Its *footprint* has two parts, in the sense of Bernstein's conditions for independence of operations. The *write-set* (the consumed resources) is the triple

$$\partial r := \{ \text{intro}_p(r), \text{intro}_e(r), \text{cell}(r) \},$$

the two introduction-occurrences $K_p(I, C_p)$, $K_e(J, C_e)$ and the head cell $\text{cell}(r) = (L, s)$ that the firing spends. The *read-set* $\partial^r r$ is the structural context on which the rule's applicability depends but which it does *not* consume: the surfaces I, J inspected by the guard $\text{near}(I, J)$, the located stack's identity L , and — in calculi where a step's enabling depends on surrounding structure, such as the boundary an ambient capability reads or the context in which an **open** acts — that surrounding structure. The contractum and any tokens the firing pushes are *produced*, part of neither set.

Lemma 3.3 (Forcing is gauge-invariant; quotient Q1). *If $t \equiv u$ and r is a redex occurrence of t , then the $\equiv$ -corresponding occurrence r of u is a redex occurrence, and $t \xrightarrow{r} t'$, $u \xrightarrow{r} u'$ imply $t' \equiv u'$. Hence $\longrightarrow$, firings, and footprints are well defined on $\equiv$ -classes.*

Proof. The gated rule (2) is stated modulo $\equiv$: its premise ($\text{near}(I, J)$ defined, head token matching) and its action (replace the cut by **compute**, pop the head) refer only to $\equiv$ -classes, since $\equiv$ leaves both the contact structure up to K 's equations and the located stack invariant. A congruence $t \equiv u$ carries the occurrence r to a $\equiv$ -equal occurrence with identical premise and identical contractum up to $\equiv$. Confluence of $\equiv$ with $\longrightarrow$ on the spent material then gives $t' \equiv u'$. $\square$

Definition 3.4 (Independence / concurrency at a configuration). Two redex occurrences r_1, r_2 co-enabled at t are *independent* (or *concurrent*), written $r_1 \text{ co } r_2$, when they satisfy the *Bernstein conditions*: no write–write overlap and no read–write overlap, while read–read overlap is permitted:

$$\partial r_1 \cap \partial r_2 = \emptyset, \quad \partial r_1 \cap \partial^r r_2 = \emptyset, \quad \partial^r r_1 \cap \partial r_2 = \emptyset.$$

That is, neither spends a resource the other spends, and neither spends a resource the other depends on. The relation co is irreflexive (a redex shares its whole write-set with itself) and symmetric.

A co-enabled pair therefore falls into exactly one of two cases, and the distinction is the whole point. Either it is independent in the Bernstein sense above — and then, by Lemma 3.5, the two orders *converge*: this is genuine concurrency, not a race, since the outcome is the same whichever fires first. Or the conditions fail, in one of two ways, each of which is a dependency, not a coincidence to be smoothed over:

- **write–write clash** ($\partial r_1 \cap \partial r_2 \neq \emptyset$): both would consume the *same* resource — the same located-stack head, or the same introduction. At one configuration the stack has a single head, so two redexes both poised to pop it are *competing*; firing either consumes the resource and destroys the other redex. This is *conflict*, a race (Section 4, relation $\#$), and the diamond does not apply to it.
- **read–write clash** ($\partial r_1 \cap \partial^r r_2 \neq \emptyset$ or symmetrically): one firing rewrites structural context the other reads — an **open** dissolving a boundary the other step is standing inside. Firing the writer changes whether or how the reader fires, so the two are ordered, not interchangeable.

Causal order $\leq$ between distinct events (Section 4) is a third, separate phenomenon: it relates an event to a *later* one that consumes or reads what the first *produced*, across configurations — not a clash between two redexes co-enabled at one configuration. The diamond below speaks only to the first, independent case.

Lemma 3.5 (Diamond for independent redexes). *If $r_1 \text{ co } r_2$ at t , then there is a configuration t_{12} with*

$$t \xrightarrow{r_1} t_1 \xrightarrow{r_2} t_{12} \quad \text{and} \quad t \xrightarrow{r_2} t_2 \xrightarrow{r_1} t_{12},$$

and the write-sets $\partial r_1, \partial r_2$ are unchanged along both paths. That is, $\longrightarrow$ is locally confluent on independent redexes. (This is not a confluence claim about $\longrightarrow$ in general; see Remark 3.6.)

Proof. By Property 2.3, firing r_1 alters only the material in ∂r_1 and its contractum, leaving everything disjoint from ∂r_1 unchanged up to $\equiv$. Since $\partial r_2 \cap \partial r_1 = \emptyset$ (no write–write overlap), both introductions of r_2 and the head cell of r_2 persist in t_1 exactly as in t . Since $\partial r_1 \cap \partial^r r_2 = \emptyset$ (no read–write overlap), r_1 alters nothing in r_2 ’s read-set: the surfaces I_2, J_2 the guard $\text{near}(I_2, J_2) = L_2$ inspects, the stack identity L_2 , and any surrounding structure r_2 depends on are untouched, so r_2 ’s guard still holds. The head token of the stack at L_2 is untouched because r_1 spends a different cell (independent co-enabled redexes on a common stack would both target its single head, violating $\partial r_1 \cap \partial r_2 = \emptyset$). Hence r_2 is enabled at t_1 , with the same write-set. Symmetrically r_1 is enabled at t_2 .

For confluence, observe that the two firings rewrite disjoint subterms of t (their write-sets are disjoint, their reads are not written by the other, and by Property 2.3 their effects are confined to those write-sets plus fresh contracta). Rewriting disjoint, non-overlapping subterms commutes: the result of performing both, in either order, is the term obtained from t by simultaneously replacing each cut instance by its contractum and each spent head cell $s_i :: S'_i$ by its tail S'_i . The two contracta are independent of each other (each is a function only of its own footprint), and the two stack pops act on distinct cells, so the outcomes coincide up to $\equiv$. This common outcome is t_{12} . $\square$

Remark 3.6 (The calculus is not confluent; all nondeterminism is conflict). Lemma 3.5 must not be misread as asserting that $\mathfrak{C}(G)$, or rho, π , the ambient calculus, or a general ciGSLT, is confluent. None of them is, and the construction neither claims nor needs global confluence. The diamond is restricted to *independent* redexes (Definition 3.4), and that restriction is exactly the definition of concurrency, not a determinism hypothesis. A co-enabled pair that is *not* independent — a write–write clash — is a genuine race, and the diamond is silent about it: the two firings consume the same resource, each destroying the other, and the run forks. All of the calculus’s nondeterminism is carried by the conflict relation $\#$ of the event structure (Section 4), never by **co**; trace equivalence (Definition 4.1) permutes only independent steps and so never identifies two resolutions of a race — they are distinct events, distinct maximal configurations, distinct causal sets, distinct spacetimes. Choosing among them is the content of Section 11.

The canonical races are write–write clashes on a shared consumed resource. Two sodium atoms competing for one chlorine: the Cl introduction lies in both write-sets, $\partial r_{\text{Na}_1} \cap \partial r_{\text{Na}_2} \ni \text{Cl} \neq \emptyset$, so the pair is *not* independent; firing either consumes the Cl and annihilates the other redex, yielding two conflicting events $\text{Na}_1\text{--Cl}$ and $\text{Na}_2\text{--Cl}$ and two configurations. Two sperm and one egg, or two cores racing to write one cache line or take one lock, have the identical shape: a shared write-set, hence $\#$. By contrast two writes to *different* locations have disjoint write-sets, genuinely commute, and are not a race at all — the only case the diamond touches. So the diamond is precisely the criterion that tells races from non-races: a race is a co-enabled pair for which the diamond fails.

There is a deeper reason the split lands here. Spacelike-separated events must have an order-independent joint outcome, or two observers who legitimately disagree about which fired first would disagree about who won, breaking frame-independence (the remark after Lemma 4.6). Spacelike pairs are therefore *forced* to commute — the diamond on **co** is the discrete shadow of Lorentz invariance of outcomes at spacelike separation — so a race cannot be spacelike. It must be a conflict, and conflict is local: it lives at a single shared resource both competitors can reach (the Cl, the egg, the cache line), a locus of decision in causal contact with both. This is the sense in which races sit “in the same light cone”: nondeterminism requires causal contact, and causal contact through a shared consumed resource is $\#$, not **co**. The construction puts races exactly where relativity says they belong.

Finally, a representational caveat. Prime event structures suffice for all three races above because each individuates its winner: the events $\text{Na}_1\text{--Cl}$ and $\text{Na}_2\text{--Cl}$ are distinct, each with its own causal past. What prime event structures do *not* capture is genuine *OR-causality* — a single event with alternative, mutually sufficient enabling sets, as when one abstracts “the egg is fertilised” away from *which* sperm did it. That requires general (non-prime) event structures [10]. We work with prime event structures throughout because the cost discipline individuates each forcing by the cell it spends, so the winner is always named.

Remark 3.7 (Why disjoint footprints, not disjoint occurrences). The footprint, not the syntactic position, is the right notion of independence under a non-free K . When K is associative–commutative (the rho case), two introductions floating in the same bag are positionally adjacent to every stack, so positional disjointness is vacuous; but they may still spend distinct located cells and hold distinct surfaces, hence be genuinely concurrent. Conversely under a free K (λ) the footprint disjointness reduces to disjointness of the rigid contexts. The footprint reading is the one that survives the whole migration spectrum of [1], because it is keyed to what is *consumed*, which is what the cost construction was built to track.

4 The event structure of a computation

We now pass from firings-in-context to *events*: firings identified across the gauge $\equiv$ (Q1) and across the reordering of concurrent steps (Q2). The standard machinery is that of asynchronous transition systems, Mazurkiewicz traces, and prime event structures [8, 9, 10, 11, 12]; we construct the event structure directly from runs and prove its axioms, invoking the external theory only for the conceptual packaging.

Definition 4.1 (Runs, trace equivalence). A *run* from t_0 is a finite sequence $\rho = (t_0 \xrightarrow{r_1} t_1 \xrightarrow{r_2} \dots \xrightarrow{r_n} t_n)$ of firings (each $t_i \in \text{Reach}(t_0)$, taken modulo $\equiv$). Its length is $|\rho| = n$. *Trace equivalence* $\sim$ is the least equivalence on runs from t_0 closed under the transposition of adjacent independent steps: if $r_i \text{ co } r_{i+1}$ at t_{i-1} , then by Lemma 3.5

$$\dots t_{i-1} \xrightarrow{r_i} t_i \xrightarrow{r_{i+1}} t_{i+1} \dots \sim \dots t_{i-1} \xrightarrow{r_{i+1}} t'_i \xrightarrow{r_i} t_{i+1} \dots$$

A *history* is a $\sim$ -class of runs; $\text{Hist}(t_0)$ is the set of histories, partially ordered by the trace-prefix order $h \leq h'$ iff some representative of h is a prefix of some representative of h' .

Lemma 4.2 (Length is a trace invariant). *If $\rho \sim \rho'$ then $|\rho| = |\rho'|$. Hence $|h| := |\rho|$ for any $\rho \in h$ is well defined.*

Proof. A generating transposition permutes two steps, changing neither the number of steps nor the multiset of firings performed. The general case follows by induction on the number of transpositions relating ρ to ρ' . $\square$

Definition 4.3 (Events). A history $p \in \text{Hist}(t_0)$ is *prime* if it has a greatest proper sub-history: a unique $p^- \leq p$ with $p^- \neq p$ such that $q \leq p$, $q \neq p \Rightarrow q \leq p^-$. The set of events is $E = E(t_0) := \{p \in \text{Hist}(t_0) : p \text{ prime}\}$, ordered by restriction of $\leq$, with the conflict relation

$$e \# e' \quad := \quad e, e' \text{ have no common upper bound in } \text{Hist}(t_0).$$

We write $\mathbb{E}(t_0) = (E, \leq, \#)$.

Intuitively a prime history is a run-modulo-reordering whose final step cannot be deferred: it is one firing together with exactly the causal past it requires. The event *is* that minimal history; e^- removes its final firing.

Lemma 4.4 (Prime-event-structure axioms; quotient Q2). $\mathbb{E}(t_0)$ is a *prime event structure*:

- (i) **Finite causes.** For every $e \in E$ the down-set $\downarrow e = \{e' \in E : e' \leq e\}$ is finite, with $|\downarrow e| = |e|$.
- (ii) **Conflict is irreflexive and symmetric.**
- (iii) **Conflict is hereditary:** $e \# e' \leq e'' \Rightarrow e \# e''$.

Proof. (i) Each prime $e' \leq e$ is determined by a firing occurring in the history e , namely the final firing of e' ; distinct primes below e end in distinct firing-occurrences of e , and conversely each of the $|e|$ firing-occurrences in e determines a unique prime below e (its minimal causal closure). Hence $e' \mapsto (\text{final firing})$ is a bijection $\downarrow e \rightarrow \{\text{firings of } e\}$, so $|\downarrow e| = |e|$, which is finite because runs are finite.

(ii) Symmetry is immediate from the symmetric phrasing. For irreflexivity, $e \leq e$, so e is a common upper bound of e with itself; thus $\neg(e \# e)$.

(iii) Suppose $e \# e'$ and $e' \leq e''$, but for contradiction e, e'' have a common upper bound u . Then $e \leq u$ and $e' \leq e'' \leq u$, so u bounds both e and e' , contradicting $e \# e'$. Hence $e \# e''$. $\square$

Definition 4.5 (Configurations). For a history $h \in \text{Hist}(t_0)$ put $[h] := \{e \in E : e \leq h\}$. A subset $x \subseteq E$ is a *configuration* when it is *down-closed* ($e' \leq e \in x \Rightarrow e' \in x$) and *conflict-free* ($e, e' \in x \Rightarrow \neg(e \# e')$). The set of configurations is $\text{Conf}(\mathbb{E}(t_0))$, ordered by inclusion.

Lemma 4.6 (Histories are configurations). *For each $h \in \text{Hist}(t_0)$, $[h]$ is a configuration with $|[h]| = |h|$. The assignment $h \mapsto [h]$ is an order-isomorphism $\text{Hist}(t_0) \cong \text{Conf}(\mathbb{E}(t_0))$.*

Proof. Down-closure: if $e' \leq e \leq h$ then $e' \leq h$, so $e' \in [h]$. Conflict-freeness: any $e, e' \leq h$ have the common upper bound h , so $\neg(e \# e')$. The cardinality $|[h]| = |h|$ follows as in Lemma 4.4(i): the primes below h are in bijection with the firing-occurrences of h , which number $|h|$ by Lemma 4.2. That $h \mapsto [h]$ is a monotone bijection with monotone inverse is the prime-algebraicity of the trace-history domain; this is the standard representation of Mazurkiewicz-trace languages and asynchronous transition systems by prime event structures and prime algebraic domains [8, 9, 10, 11], whose hypotheses (the diamond property and finiteness, here Lemmas 3.5 and 4.2) we have verified for $\mathfrak{C}(G)$. $\square$

Remark 4.7 (Frame-independence). Lemma 4.6 is the computational content of relativistic frame-independence. A run is a choice of total order on its firings — a choice of observer’s clock. Trace equivalence quotients runs by the reordering of concurrent (spacelike) steps, exactly as different inertial observers disagree on the time-order of spacelike-separated events. The invariant is the partial order itself: $\mathbb{E}(t_0)$ is the quotient of all linearisations by permutation of the independent ones, and a configuration is the frame-free history that all observers agree on.

5 Configurations are causal sets

Definition 5.1 (Causal set; [2]). A *causal set* is a partial order $(C, \preceq)$ that is *locally finite*: for all $x \preceq y$ the order interval

$$[x, y] := \{z \in C : x \preceq z \preceq y\}$$

(the discrete *causal diamond*) is finite. A *measured causal set* is a causal set with the counting measure $\mu(A) = |A|$.

Theorem 5.2 (Every configuration is a causal set). *For each $x \in \text{Conf}(\mathbb{E}(t_0))$, the pair $(x, \leq|_x)$ is a causal set under the inherited order, with $\mu = |\cdot|$. In particular each history h yields a measured causal set $[h]$, and the maximal configurations — the complete runs modulo reordering — are the inextendible discrete spacetimes of the computation.*

Proof. The inherited relation is a partial order. For local finiteness, take $e \leq e'$ in x ; then $[e, e'] \subseteq \downarrow e'$, which is finite by Lemma 4.4(i). Thus $[e, e']$ is finite. (Note this holds even when x is infinite, i.e. when t_0 admits non-terminating runs: the causal set may be infinite, but every causal diamond in it is finite, which is exactly local finiteness.) $\square$

The theorem is the structural half of *Order + Number = Geometry*: the order is in hand, and it is of the right kind. The next section supplies the number.

6 Number is volume: spent phlogiston is spacetime volume

In causal-set theory the continuum spacetime volume of a region is recovered, in the manifoldlike limit, as the number of causet elements it contains: volume is cardinality [2, 3]. We show that on $\mathbb{E}(t_0)$ this cardinality is the spent phlogiston, so that the modulus of [1] is the volume measure.

Notation 6.1. For an event e , let $\text{phlo}(e)$ denote the phlogiston spent to reach e : the number of stack cells popped along any run-prefix whose history is e . By Lemma 4.2 this is independent of the chosen linearisation.

Theorem 6.2 (Number = Volume = phlogiston). *For every $e \in E(t_0)$,*

$$\mu(\downarrow e) = |\downarrow e| = |e| = \text{phlo}(e).$$

Consequently, for $e \leq e'$ the volume of the causal diamond is the phlogiston spent strictly within it,

$$\mu([e, e']) = |[e, e']| = \text{phlo}(e') - \text{phlo}(e) + 1,$$

counting both endpoints; and the volume of the causal past of any region is the total phlogiston its computation has burned.

Proof. The first two equalities are Lemma 4.4(i). For $|e| = \text{phlo}(e)$: a run-prefix realising the history e performs exactly $|e|$ firings (Lemma 4.2), and by the modulus Property 2.2 each firing is a forcing that pops exactly one stack cell, the number of cells popped equalling the number of forced redexes. Hence the cells popped number $|e|$, i.e. $\text{phlo}(e) = |e|$. The diamond formula follows since $[e, e'] = \downarrow e' \setminus \downarrow e^\circ$ where $\downarrow e^\circ = \downarrow e \setminus \{e\}$ is the strict past, so $|[e, e']| = |\downarrow e'| - |\downarrow e| + 1 = \text{phlo}(e') - \text{phlo}(e) + 1$. $\square$

Remark 6.3 (Phlogiston as a volume, made literal). The closing correspondence of [1] read phlogiston as a potential on the transition graph. Theorem 6.2 gives it the complementary geometric reading demanded by causal-set theory: spent phlogiston is the four-volume swept out by the computation's causal past. The temporal stack is therefore not merely a clock but the *volume measure* of the emergent spacetime — and because the metering is lazy (charged at forcing, not at exposure), the measure is accumulated exactly in step with the spacetime's own growth. Burning phlo is making spacetime.

7 Proper time and the Lorentzian signature

A causal set is Lorentzian, not Riemannian: its order is the causal (lightcone) order, and the geometric magnitude along a timelike pair is proper time, characterised in the continuum by the *reverse* triangle inequality (timelike geodesics maximise proper time). In causal-set theory the discrete proper time between two related elements is the length of the longest chain between them [3]. We adopt this and prove the inequality holds here; we then show that located authority is exactly what produces a nondegenerate signature.

Definition 7.1 (Chains, antichains, proper time). A *chain* in a configuration x is a set of pairwise $\leq$ -comparable events (a discrete timelike worldline); an *antichain* is a set of pairwise incomparable events (a discrete spacelike slice). For $e \leq e'$ the *proper time* is

$$\tau(e, e') := \max\{k : e = e_0 < e_1 < \dots < e_k = e'\},$$

the length of the longest chain in $[e, e']$ (finite by local finiteness, Theorem 5.2).

Proposition 7.2 (Lorentzian reverse triangle inequality). *For $e \leq e' \leq e''$ in any configuration,*

$$\tau(e, e'') \geq \tau(e, e') + \tau(e', e'').$$

Proof. Let C_1 be a longest chain from e to e' (length $\tau(e, e')$) and C_2 a longest chain from e' to e'' (length $\tau(e', e'')$). Their concatenation at the shared element e' is a chain from e to e'' of length $\tau(e, e') + \tau(e', e'')$. The longest such chain is at least this long, giving the inequality. $\square$

This is the signature of time: unlike a metric, τ adds *against* subdivision, so inserting an intermediate event can only lengthen proper time, never shorten it — the discrete analogue of the twin paradox. The spatial direction, by contrast, is carried by antichains, on which τ is undefined (incomparable elements have no chain between them). The question is whether a configuration has *both*. The answer is governed, as everywhere in [1], by the equational theory of K through the located stack.

Proposition 7.3 (Signature from location). *Fix a configuration $x \in \text{Conf}(\mathbb{E}(t_0))$.*

- (a) (Time from $::$.) *The firings in x that draw on a single located stack $S(L, -)$ form a chain: they are pairwise $\leq$ -comparable, ordered by the depth in L at which each pops.*
- (b) (Space from K .) *If $e_1, e_2 \in x$ draw on distinct located stacks and hold surfaces I_1, I_2 with $\text{near}(I_1, I_2)$ and $\text{near}(I_2, I_1)$ undefined, and neither produces an introduction the other consumes, then e_1, e_2 are concurrent: they form an antichain.*
- (c) (Signature.) *If x contains a located stack drawn on at least twice and two such non-near surfaces drawn concurrently, then x is neither a chain nor an antichain: its ordering fraction (Definition 7.5) lies strictly between 0 and 1, the combinatorial mark of a $(1+n)$ -dimensional causal structure.*

Proof. (a) Two firings spending head cells of the same stack $S(L, -)$ cannot be independent, since the head cell is a single shared resource: at most one of them is enabled to pop the current head at any configuration (Definition 3.4 excludes a common cell from co). Concretely, if the stack is $s_1 :: s_2 :: \dots$, then s_2 becomes the head only after a firing pops s_1 ; a firing popping s_2 therefore has, in its causal past, a firing popping s_1 , so the former $\leq$ relates the latter. By induction the draws on L are linearly ordered by stack depth, hence a chain.

(b) The hypotheses say precisely $\partial e_1 \cap \partial e_2 = \emptyset$: distinct stacks give distinct cells, the non-nearness of the surfaces means neither's surface is the other's (no shared introduction through surface matching), and the last clause excludes a producer/consumer link. Thus $e_1 \text{ co } e_2$. Independent events are incomparable in $\leq$ (neither lies in the other's causal past, since neither's footprint is produced by the other) and compatible (the diamond, Lemma 3.5, gives a common upper bound), so they are concurrent and lie in a common antichain.

(c) By (a) the repeated draws on the stack give a chain of length ≥ 2 , contributing at least one related pair, so the ordering fraction $r(x) > 0$ and x is not an antichain. By (b) the two concurrent non-near draws give an incomparable pair, so $r(x) < 1$ and x is not a chain. The intermediate value $0 < r(x) < 1$ is, via the Myrheim–Meyer estimator (Definition 7.5, Proposition 7.6), a finite dimension > 1 . $\square$

Remark 7.4 (The welding, restated). Proposition 7.3 is the formal content of the entangling of space and time announced in [1]. Without location, a configuration would be a bare product: the temporal stacks would order nothing across surfaces (no shared cells to chain distinct surfaces) and K would spread surfaces with no temporal coupling, giving an antichain of independent timelines — a space and a time that vary independently, i.e. a foliated $\Sigma \times \mathbb{R}$ with a global “now.” Location is what binds a temporal chain (a) *to* a spatial surface, so that a single located stack is a worldline pinned at a place, and the family of located stacks across non-near surfaces is the spacelike spread of worldlines. The mixing of (a) and (b) destroys the global foliation — there is no slice cutting every worldline at the same stack-depth, because depths on distinct stacks are incomparable — and that is what makes the structure a spacetime rather than a space times a time.

Definition 7.5 (Ordering fraction; Myrheim–Meyer dimension). For a finite configuration or finite causal diamond A with $N = |A| \geq 2$, let $R(A)$ be the number of *related* (comparable, ordered) pairs. The *ordering fraction* is

$$r(A) := \frac{R(A)}{\binom{N}{2}} = \frac{2 R(A)}{N(N-1)} \in [0, 1].$$

The *Myrheim–Meyer dimension* $d_{\text{MM}}(A)$ is the unique real $d \geq 1$ solving the sprinkling relation [4, 5]

$$r = \frac{\Gamma(d+1) \Gamma(\frac{d}{2})}{4 \Gamma(\frac{3d}{2})}, \quad (3)$$

the expected ordering fraction of a Poisson sprinkling into a causal diamond of d -dimensional Minkowski space.

Proposition 7.6 (Extremal bounds). *The right-hand side of (3) is strictly decreasing in d on $[1, \infty)$, with value 1 at $d = 1$ and limit 0 as $d \rightarrow \infty$. Hence $r(A) = 1$ iff A is a chain, forcing $d_{\text{MM}} = 1$ (pure time, no space); $r(A) = 0$ iff A is an antichain, the degenerate $d_{\text{MM}} = \infty$ limit (pure space, no time); and any A satisfying the hypotheses of Proposition 7.3(c) has $0 < r(A) < 1$, hence finite $d_{\text{MM}} > 1$.*

Proof. $r(A) = 1$ means every pair is comparable, i.e. A is a chain; $r(A) = 0$ means no pair is comparable, i.e. A is an antichain. At $d = 1$ the diamond of 1-dimensional Minkowski space is an interval of the line, whose every sprinkled pair is timelike-related, giving expected $r = 1$; substitution $d = 1$ into (3) gives $\Gamma(2)\Gamma(1/2)/(4\Gamma(3/2)) = 1 \cdot \sqrt{\pi}/(4 \cdot \frac{\sqrt{\pi}}{2}) = 1$. Monotonicity and the $d \rightarrow \infty$ limit are the standard properties of (3) [5]; we do not reproduce the asymptotic analysis of the Gamma-quotient here. The final clause is Proposition 7.3(c). $\square$

8 The functor and the spacetime invariant

We now assemble the map and identify the equivalence it inverts — which is *not* ordinary bisimulation, but its concurrency-aware refinement.

Definition 8.1 (The category of measured causal sets). **MeasCaus** has as objects measured causal sets $(C, \preceq, \mu)$ with μ the counting measure, and as morphisms the *rigid measure-preserving causal maps*: injections $g : C \rightarrow C'$ that are order-embeddings ($x \preceq y \iff g(x) \preceq' g(y)$) whose image is down-closed and which preserve covers ($x \triangleleft y \iff g(x) \triangleleft' g(y)$). Down-closed image and order-embedding force $\mu([x, y]) = \mu'([g(x), g(y)])$, so g is automatically measure-preserving on intervals. Isomorphisms are the order-and-measure isomorphisms.

The object map sends a pointed system to its event structure, regarded as the domain of its measured configurations.

Definition 8.2 (The map on objects). $\mathcal{M}(\mathfrak{C}(G), t_0) := \mathbb{E}(t_0) = (E, \leq, \#)$ with counting measure, equivalently its domain of measured configurations $\text{Conf}(\mathbb{E}(t_0))$. A single discrete spacetime is a maximal configuration $(x, \leq | x, | \cdot |) \in \mathbf{MeasCaus}$ (Theorem 5.2).

To make $\mathcal{M}$ a functor we must say which **Cost-ciGSLT**-morphisms it sends to causal maps, and here a subtlety of concurrency theory is decisive. Ordinary (interleaving) bisimulation equates systems with the same tree of possible step-sequences; it is blind to whether two enabled steps are

concurrent or in conflict, because both present as a branch. But concurrency is exactly the space in our spacetime: an antichain and a chain can have the same interleaving tree. Hence the spacetime cannot be an invariant of interleaving bisimulation. The right notion is the one that additionally matches the independence relation along the bisimulation: *hereditary history-preserving bisimulation* (HHPB), which van Glabbeek and Goltz [13] identify as precisely the behavioural equivalence corresponding to isomorphism of event structures.

Definition 8.3 (Quote-faithful HHP-morphism). A **Cost-ciGSLT_•**-morphism $f : (\mathfrak{C}(G), t_0) \rightarrow (\mathfrak{C}(G'), t'_0)$ is *footprint-preserving* when it is quote-faithful (preserves signatures and surfaces, hence the definedness of **near** and the read-set $\partial^r r$) and sends each firing r to a firing $f(r)$ with $\partial f(r) = f(\partial r)$ and $\partial^r f(r) = f(\partial^r r)$, bijectively on both consumed resources (the write-set) and depended-on context (the read-set). It is an *HHP-morphism* when, in addition, the induced relation on runs is a history-preserving bisimulation: it respects enabling and independence, and back-and-forth along already-matched causal pasts. Write **Cost-ciGSLT_•^{hhp}** for the subcategory of **Cost-ciGSLT_•** on these morphisms.

Lemma 8.4 (HHP-morphisms preserve the causal structure). *A footprint-preserving HHP-morphism f induces a map $\mathbb{E}(f) : \mathbb{E}(t_0) \rightarrow \mathbb{E}(t'_0)$ on events that preserves and reflects $\leq$, preserves and reflects $\#$, and preserves the counting measure: $\mu(\downarrow e) = \mu'(\downarrow \mathbb{E}(f)(e))$.*

Proof. Because f is footprint-preserving, it carries co-enabled redex occurrences to co-enabled occurrences with $\partial f(r) = f(\partial r)$ and $\partial^r f(r) = f(\partial^r r)$, and bijectivity on both sets carries each Bernstein clause to its image: write–write, read–write, and write–read overlaps are preserved and reflected. Hence $r_1 \text{ cor } r_2 \iff f(r_1) \text{ co } f(r_2)$, so f preserves and reflects independence (and likewise the failures of independence, i.e. races). Preserving enabling and independence, f sends runs to runs and trace-equivalent runs to trace-equivalent runs (it commutes with the generating transpositions), so it descends to $\text{Hist}(t_0) \rightarrow \text{Hist}(t'_0)$, monotone for the prefix order, carrying primes to primes; this is $\mathbb{E}(f)$. Reflection of $\leq$ and the back-condition of an HHP-bisimulation give injectivity and order-reflection. Conflict is preserved and reflected because f preserves and reflects the existence of common upper bounds (it is a bijection on the matched configuration lattices, by the hereditary back-and-forth). Finally, by Lemma 4.4(i) and Lemma 4.2, $\mu(\downarrow e) = |e|$ counts firings, and f is a bijection on the firings of e onto those of $f(e)$, so the measure is preserved. $\square$

Theorem 8.5 (Functoriality). $\mathcal{M} : \mathbf{Cost-ciGSLT}_{\bullet}^{\text{hhp}} \rightarrow \mathbf{MeasCaus}$, sending $(\mathfrak{C}(G), t_0) \mapsto \mathbb{E}(t_0)$ and $f \mapsto \mathbb{E}(f)$, is a functor.

Proof. $\mathbb{E}(f)$ is a rigid measure-preserving causal map by Lemma 8.4 (order-embedding with down-closed image from preservation/reflection of $\leq$ and the back-condition; cover preservation from preservation of the immediate-enabling step). Identities: $\mathcal{M}(\text{id}) = \mathbb{E}(\text{id})$ fixes every run, hence every history and every event, so $\mathbb{E}(\text{id}) = \text{id}$. Composition: for composable f, g , the induced action on runs satisfies $\mathbb{E}(g \circ f) = \mathbb{E}(g) \circ \mathbb{E}(f)$ since both are computed stepwise on firings and $(g \circ f)(\partial r) = g(f(\partial r))$. Hence $\mathcal{M}$ preserves identities and composition. $\square$

Corollary 8.6 (The spacetime is an HHPB-invariant; quotient Q3). *If $(\mathfrak{C}(G), t_0)$ and $(\mathfrak{C}(G'), t'_0)$ are related by a quote-faithful hereditary history-preserving bisimulation, then $\mathbb{E}(t_0) \cong \mathbb{E}(t'_0)$ as measured causal sets, and hence their domains of measured configurations are isomorphic. In particular the discrete spacetime of a cost-accounted computation is well defined independently of the presentation, up to HHPB.*

Proof. A quote-faithful HHPB is an HHP-morphism with an HHP-inverse; by Theorem 8.5 both induce causal maps, and functoriality makes the induced maps mutually inverse isomorphisms in

MeasCaus, by van Glabbeek–Goltz’s correspondence between HHPB and event-structure isomorphism [13]. $\square$

Remark 8.7 (Interleaving bisimulation is the wrong invariant, and why it matters). It is worth stating the negative result plainly, since it corrects a natural expectation. The **ciGSLT**-morphisms of [1] are defined up to bisimulation; but *interleaving* bisimulation identifies a chain of two firings with a confluent diamond of two concurrent firings whenever their interleaving trees coincide, and these have different causal sets — a 2-chain (timelike) versus a 2-antichain (spacelike). Were the spacetime an interleaving-bisimulation invariant, time and space would be indistinguishable. They are distinguished exactly by the independence relation, which the footprint makes available and which HHPB preserves. The located stack is therefore doing double duty: operationally it prevents ambient authority ([1], Section 11), and semantically it is the carrier of the independence relation that makes the spacetime invariant well defined. The ocap discipline and the Lorentzian structure are the same structure seen twice.

9 The field-equation shape: naturality of the cover

The general-relativity correspondence of [1] was a two-way coupling: the information distribution shapes the nearness structure, and the nearness structure selects the next redistribution. We render the coupling as a property of the configuration domain — that one reduction step is one cover, whose new geometry is a *local functional* of the information consumed.

Proposition 9.1 (One step is one cover; the increment of geometry is local). *Let $x \leq x'$ be a cover in $\text{Conf}(\mathbb{E}(t_0))$ (so $x' = x \cup \{e\}$ for a single new event e , by Lemma 4.6 and Lemma 4.2). Then:*

- (i) $x \leq x'$ corresponds to exactly one firing r with $e =$ the prime generated by r in the history x' ;
- (ii) the new causal links into e — the events $e'' \leq e$ — are precisely the events of x that produced the introductions and exposed the head cell in ∂r ; in particular the link-set of e is a bounded functional of ∂r alone (at most two producer-events for the introductions and one for the cell);
- (iii) the firings enabled at x' but not at x are exactly those whose footprint is met by the contractum of r (its produced introductions and pushed cells).

Hence “geometry sourced by the distribution” is (ii) — the local information consumed determines the new event’s lightcone links — and “distribution guided by geometry” is (iii) — the new local geometry determines which interactions come next. The map “perform one reduction” $\mapsto$ “add one measured cover” is natural in $\longrightarrow$.

Proof. (i) Adding one firing to a history lengthens it by one (Lemma 4.2); by Lemma 4.6 the corresponding configurations differ by one event, the prime whose final firing is r . (ii) By definition $e' \leq e$ iff $e' < e$ with nothing strictly between; $e' < e$ holds iff e' lies in the causal past required to enable r , which by Property 2.3 and Definition 3.2 means e' produced a member of ∂r (an introduction r contracts, or the cell r pops); immediacy selects the most recent such producer per resource. There are at most three resources in ∂r , bounding the link-set. (iii) A firing q is newly enabled at x' iff its footprint, absent or incomplete at x , is completed by what r produced; by Property 2.3 the only new material is the contractum of r , so q ’s footprint must meet it. Naturality is the statement that the assignment commutes with composition of reduction steps, immediate from (i)–(iii) applied stepwise. $\square$

Remark 9.2 (Discrete Einstein coupling). Proposition 9.1 is the discrete, computational shadow of $G_{\mu\nu} = 8\pi T_{\mu\nu}$ read as Wheeler’s two-way slogan: *matter tells spacetime how to curve; spacetime tells matter how to move*. Here the consumed information (the footprint, the local stress) is T ; the new event’s link-set (the local causal geometry it creates) is the local increment of the metric; and (ii)–(iii) are the two directions of the coupling, each *local and bounded*, as a field equation demands. We do not claim a quantitative field equation — that would require the manifoldlike limit and a continuum action, and is the open problem flagged in [1]. What is proved is the *shape*: the geometry-information coupling is a local functional, step-by-step, exactly as a hyperbolic field equation is.

10 The four quotients and the manifold

We can now name precisely the senses in which the target is a *quotiented* spacetime. The construction performs four identifications, the first three of which deliver the causal set as an invariant and the fourth of which delivers the continuum manifold.

- (Q1) **Gauge: structural congruence** $\equiv$. Re-bracketing the bag, α -renaming, unit insertions — congruence steps consume no token and leave the stack invariant. They move the description, not the events. These are the coordinate changes of the discrete spacetime; firings act on $\equiv$ -classes (Lemma 3.3).
- (Q2) **Frame: concurrent permutation (trace equivalence)**. Runs differing only by the order of concurrent (spacelike) firings are the same history (Definition 4.1); the quotient of all linearisations by permutation of the independent ones is the causal partial order (Lemma 4.6). This is relativistic frame-independence.
- (Q3) **Presentation: hereditary history-preserving bisimulation**. Quote-faithful HHPB-equivalent systems have isomorphic measured causal sets (Corollary 8.6); this is the level on which $\mathcal{M}$ is a functor, and it is strictly finer than interleaving bisimulation, which would collapse the space.
- (Q4) **Continuum: the Hauptvermutung class**. A manifoldlike causal set is identified with the family of Lorentzian manifolds into which it faithfully embeds, unique up to approximate isometry (Definition 10.1 below).

Definition 10.1 (Faithful embedding; manifoldlike; Hauptvermutung quotient; [2, 6, 7]). A causal set $(C, \preceq)$ *faithfully embeds* into a Lorentzian manifold (M, g) at density ϱ when there is an order-preserving injection $\iota : C \hookrightarrow M$ (relating $\preceq$ to the causal order of g) such that C is, with high probability, a Poisson sprinkling of M at density ϱ — equivalently, the counting measure approximates ϱ times the volume measure on regions large compared with ϱ^{-1} . C is *manifoldlike* when such an (M, g) exists. The *Hauptvermutung* of causal-set theory asserts that the manifold is then unique up to approximate isometry; we define

$$\text{Manifold}(C) := [(M, g)]_{\approx},$$

the approximate-isometry class, when C is manifoldlike, and undefined otherwise. A precise closeness measure realising $\approx$ is Noldus’s Lorentzian Gromov–Hausdorff distance [6], and the recovery of continuum topology and dimension is treated in [7, 5, 4].

Definition 10.2 (The full map). For $(\mathfrak{C}(G), t_0) \in \mathbf{Cost}\text{-}\mathbf{ciGSLT}_{\bullet}^{\text{hhp}}$ and a chosen maximal configuration x (one consistent history; see Section 11 on the choice), the spacetime map is the composite

$$(\mathfrak{C}(G), t_0) \xrightarrow{\mathcal{M}} \mathbb{E}(t_0) \xrightarrow{\text{select } x} (x, \leq x, | \cdot |) \xrightarrow{[\cdot]_{\approx}} \text{Manifold}(x),$$

the last arrow defined precisely when x is manifoldlike. The “(quotiented) spacetime manifold” of the computation is $\text{Manifold}(x)$; the robust, always-defined invariant beneath it is the measured causal set $(x, \leq x, | \cdot |)$, well defined up to (Q1)–(Q3).

Remark 10.3 (The manifold is a limit, not a guarantee). We stress the honest reading. The clean theorem is $\mathcal{M} : \mathbf{Cost}\text{-}\mathbf{ciGSLT}_{\bullet}^{\text{hhp}} \rightarrow \mathbf{MeasCaus}$ (Theorem 8.5): every cost-accounted computation has a measured causal set, functorially and HHPB-invariantly. The manifold is the further *idealisation* of Definition 10.1, available only for manifoldlike configurations and even then only up to the Hauptvermutung. Most configurations of a generic $\mathfrak{C}(G)$ will not be manifoldlike — they will be “too ordered” (long chains, low dimension, from a single heavily-drawn stack) or “too disordered” (wide antichains, from many non-interacting surfaces). The interesting question, parallel to the central question of causal-set dynamics, is which $\mathfrak{C}(G)$ and which t_0 produce manifoldlike spacetimes, and with what effective dimension — which by Proposition 7.3 is a question about the equational theory of K and the discipline of location.

11 The conflict seam: one spacetime or a sum over them

Definition 10.2 selected a single maximal configuration. The selection is where conflict ($\#$) — the third relation of the event structure — enters, and it is exactly the seam between the two correspondences of [1].

A maximal *conflict-free* down-closed set of events is one consistent history: its image is a single measured causal set, a single classical spacetime. This is the *general-relativity reading*: a decohered, definite geometry. The branching of $\#$ — competition of two receivers for one located stack ([1], Section 11) — is the choice of which spacetime.

Keeping the conflict, rather than choosing through it, and weighting events in a semiring R , gives the second reading. The semiring ladder of [1] climbs from the Boolean semiring (bare nondeterminism), through the tropical semiring $(\mathbb{R} \cup \{\infty\}, \min, +)$ (cost; the tropical path-weighting refines the proper time of Section 7 from longest *chain* to longest *weighted chain*), to the non-negative reals (probabilistic), to the complex numbers (*amplitudes*). At $R = \mathbb{C}$ the object is the whole branching ensemble and the amplitude of an outcome sums over the conflicting histories reaching it — the *quantum-field reading*, in the lineage of Girard’s geometry of interaction and the categorical quantum mechanics of Abramsky and Coecke [15, 16].

Here the obstruction flagged in [1] reappears precisely and must be stated, not hidden. Interference *cancels* contributions to a shared outcome; but the configuration domain of an event structure is a *stable* family, and the equivalence we used to make the spacetime an invariant, HHPB, is a branching-time notion that forbids exactly such cancellation [13, 10]. The genuinely quantum reading therefore pulls the equivalence from branching time toward a linear-time, trace-indexed semantics — a reformulation, not a relabelling. We record this as the boundary of the present construction: the functor $\mathcal{M}$ of Theorem 8.5 lands the GR reading (one configuration, one causal set, HHPB-invariant); the QFT reading requires replacing $\mathbf{MeasCaus}$ -valued configurations by an R -weighted, linear-time object on which cancellation is admissible, and is left, as in [1], to a separate development.

12 Summary of the construction

We collect the specification in one place. The map $\mathcal{M} : \mathbf{Cost}\text{-}\mathbf{ciGSLT}_{\bullet}^{\text{hhp}} \rightarrow \mathbf{MeasCaus}$ is defined by:

Spacetime datum	Cost-accounted origin
event	a cut firing (forcing): Def. 3.2
causal order $e \leq e'$	enabling: e' consumes what e produced / exposed: §4
conflict $e \# e'$ (race)	write–write clash: competition for one consumed resource (stack head, egg, cache line): Def. 4.3, Rem. 3.6
concurrency (spacelike)	Bernstein independence (no write–write or read–write overlap) $e \text{ co } e'$: Def. 3.4
measure / volume	spent phlogiston = $ \downarrow e $: Thm. 6.2
proper time	longest chain; reverse triangle: Prop. 7.2
timelike worldline	draws on one located stack $S(L, -)$: Prop. 7.3(a)
spacelike slice	non-near surfaces drawn concurrently: Prop. 7.3(b)
$(1+n)$ signature	located authority mixing chains and antichains: Prop. 7.3(c)
field-equation coupling	link-set of new event is a local functional of ∂r : Prop. 9.1
the manifold	Hauptvermutung class of a manifoldlike configuration: Def. 10.1

and the “(quotiented)” resolves into the four identifications (Q1) $\equiv$ gauge, (Q2) trace/frame, (Q3) HHPB/presentation, (Q4) Hauptvermutung/continuum (§10). The robust output is the measured causal set, proved to exist functorially for every pointed cost-accounted theory; the manifold is its continuum idealisation when manifoldlike.

The single structural lever throughout is the one [1] isolated: the equational theory of K and its consequence, the located stack. A free K gives separable, foliated structure (independent timelines); any equation on K forces the nominal surface and the located stack, which welds a temporal chain to a spatial surface and so makes the structure a spacetime. The object-capability discipline and the Lorentzian signature are, as Section 8 shows, two faces of the same located authority.

References

- [1] L. G. Meredith, *Continued Interactive GSLTs and the Cost Endofunctor*, F1R3FLY.io, 2026. <https://github.com/F1R3FLY-io/publications/blob/main/cost-accounting-as-monad/continued-gslt-cost-v2.pdf>
- [2] L. Bombelli, J. Lee, D. Meyer, R. D. Sorkin, *Space-time as a causal set*, Physical Review Letters **59** (1987) 521–524.
- [3] R. D. Sorkin, *Causal sets: Discrete gravity*, in *Lectures on Quantum Gravity* (Valdivia 2002), A. Gomberoff and D. Marolf, eds., Springer, 2005; arXiv:gr-qc/0309009.
- [4] J. Myrheim, *Statistical geometry*, CERN preprint TH-2538, 1978.

- [5] D. A. Meyer, *The Dimension of Causal Sets*, Ph.D. thesis, Massachusetts Institute of Technology, 1988.
- [6] J. Noldus, *A Lorentzian Lipschitz, Gromov–Hausdorff notion of distance*, Classical and Quantum Gravity **21** (2004) 839–850; L. Bombelli and J. Noldus, *The moduli space of causal sets*, Classical and Quantum Gravity **21** (2004) 4429.
- [7] S. Major, D. Rideout, S. Surya, *On recovering continuum topology from a causal set*, Journal of Mathematical Physics **48** (2007) 032501.
- [8] M. Nielsen, G. Plotkin, G. Winskel, *Petri nets, event structures and domains, part I*, Theoretical Computer Science **13** (1981) 85–108.
- [9] G. Winskel, *Event structures*, in *Petri Nets: Applications and Relationships to Other Models of Concurrency*, Lecture Notes in Computer Science **255**, Springer, 1987, 325–392.
- [10] G. Winskel, M. Nielsen, *Models for concurrency*, in *Handbook of Logic in Computer Science*, vol. 4, S. Abramsky, D. Gabbay, T. Maibaum, eds., Oxford University Press, 1995, 1–148.
- [11] A. Mazurkiewicz, *Trace theory*, in *Petri Nets: Applications and Relationships to Other Models of Concurrency*, Lecture Notes in Computer Science **255**, Springer, 1987, 279–324.
- [12] M. A. Bednarczyk, *Categories of Asynchronous Systems*, Ph.D. thesis, University of Sussex, 1987.
- [13] R. J. van Glabbeek, U. Goltz, *Refinement of actions and equivalence notions for concurrent systems*, Acta Informatica **37** (2001) 229–327.
- [14] J.-J. Lévy, *Réductions correctes et optimales dans le lambda-calcul*, Thèse de doctorat d’état, Université Paris VII, 1978.
- [15] J.-Y. Girard, *Geometry of interaction I: Interpretation of System F*, in *Logic Colloquium ’88*, North-Holland, 1989, 221–260.
- [16] S. Abramsky, B. Coecke, *A categorical semantics of quantum protocols*, in *Proc. 19th IEEE Symposium on Logic in Computer Science (LICS)*, 2004, 415–425.